PRAGYA: AI Checkout Recovery Agent

Turning hesitation into conversion with real-time AI intervention

Built for

Kasparro

Agentic Commerce Hackathon

Track 2: AI Assisted Checkout Recovery

Product Documentation

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Repository:
https://github.com/chandana022005/AI-Checkout-Recovery-Agent

AI-powered Shopify-like checkout recovery system using real-time behavioural signals and contextual product recommendations.

Pragya – AI Checkout Recovery Assistant

Product Document

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1. Problem Overview

E-commerce platforms suffer from significant revenue loss due to cart abandonment, where users add products to their cart but fail to complete the checkout process.

Most existing solutions rely on delayed interventions such as reminder emails or notifications, which do not address the user's concern at the moment of hesitation. These approaches fail because they are reactive rather than proactive and lack personalization.

Users abandon checkout due to reasons such as:

- High or unexpected shipping costs
- Price hesitation
- Lack of trust or clarity
- Confusion during checkout

There is a clear need for a system that can **identify hesitation in real-time and assist users instantly before they leave**.

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2. Our Approach

We propose **Pragya**, a real-time AI-powered checkout assistant that intervenes only when necessary and helps users complete their purchase.

Instead of acting after abandonment, Pragya:

- Detects hesitation using behavioral signals
- Understands the reason behind hesitation
- Engages the user with minimal, context-aware conversation
- Suggests actionable solutions

The system is designed to be **non-intrusive**, meaning it does not interrupt users unnecessarily and only activates when a meaningful signal is detected.

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3. Key Product Decisions

Decision 1: Minimal Intervention

We chose not to show the assistant immediately.

Instead:

- Trigger only after inactivity (e.g., 20 seconds)
- Or clear hesitation signals

✓ Avoids annoying the user
✓ Feels premium

Decision 2: Conversation, Not Notification

Traditional systems send messages like:

"You forgot your cart"

We instead designed:

A real-time conversation

✓ More human
✓ More effective

Decision 3: Solve, Not Sell

Instead of pushing discounts:

- First understand user problem
- Then solve it

Example:

- If shipping concern → suggest add-on
- If price concern → explain value

✓ Preserves business value
✓ Builds trust

Decision 4: Hybrid Intelligence

We combine:

- Logic (math, thresholds)
- AI (conversation, understanding)

✓ Ensures accuracy + flexibility

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4. Core User Experience

Step 1: User enters checkout

Step 2: System observes behavior

Step 3: Detects hesitation

Step 4: AI assistant appears

Step 5: User interacts naturally

Step 6: System suggests solution

Step 7: User completes purchase

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5. Example Scenarios

Scenario 1: Shipping Concern

User: "Why is shipping ₹100?"
System:

- Calculates gap to free shipping
- Suggests small add-on

Scenario 2: Price Hesitation

User: "This is expensive"
System:

- Explains product value
- Avoids aggressive selling

Scenario 3: Idle User

User inactive for 20 seconds
System:

- Starts gentle conversation
- Offers help

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6. Trade-offs & Constraints

Trade-off 1: Simplicity vs Intelligence

We limited intervention rules to keep system fast and reliable.

Trade-off 2: No Heavy Personalization

We did not use long-term user data
→ keeps system lightweight

Trade-off 3: AI Dependence

System relies on AI for responses
→ requires careful prompt design

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7. Why This Works Better

Traditional systems:
❌ Delayed

❌ Generic
❌ Reactive

Pragya:
✅ Real-time
✅ Context-aware
✅ Proactive

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8. Success Metrics

We evaluate success based on:

- Increased checkout completion rate
- Reduced cart abandonment
- User engagement with assistant
- Number of successful interventions

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9. Future Vision

Pragya can evolve into:

- Multi-channel assistant (WhatsApp, SMS)
- Personalized recommendation engine
- Full AI commerce agent